

JOUR 8500 RLA

Millions of online book co-purchases reveal partisan differences in the consumption of science

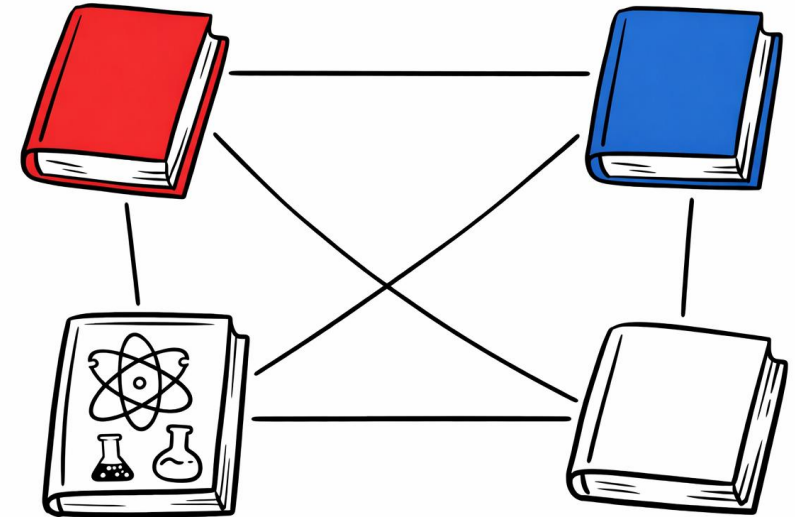
Presenter: Jikai Sun

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Shi, F., Shi, Y., Dokshin, F. A., Evans, J. A., & Macy, M. W. (2017). Millions of online book co-purchases reveal partisan differences in the consumption of science. *Nature Human Behaviour*, 1(4), 0079.

Background

- **Science bridges partisan divide** (Survey performed by AAAS and the Pew Research Center)
- Instead of using questionnaire survey, the study evaluate science preference through **co-purchases** of political books and science books.



Data and network

- **Online data** – Amazon and Barnes & Noble
- **Books** – categorized into political books, science books, and non-science books
- **Political books** – conservative (673), liberal (583)
- **Science books** – 27 disciplines (humanities, physical sciences, life sciences, social sciences)
- **Non-science books** – four categories (arts, sports, literature, religion)

Results

- (a) books are more likely to be purchased together with **books of the same color**
- (b) readers of blue books have broader disciplinary interest

FIG 1

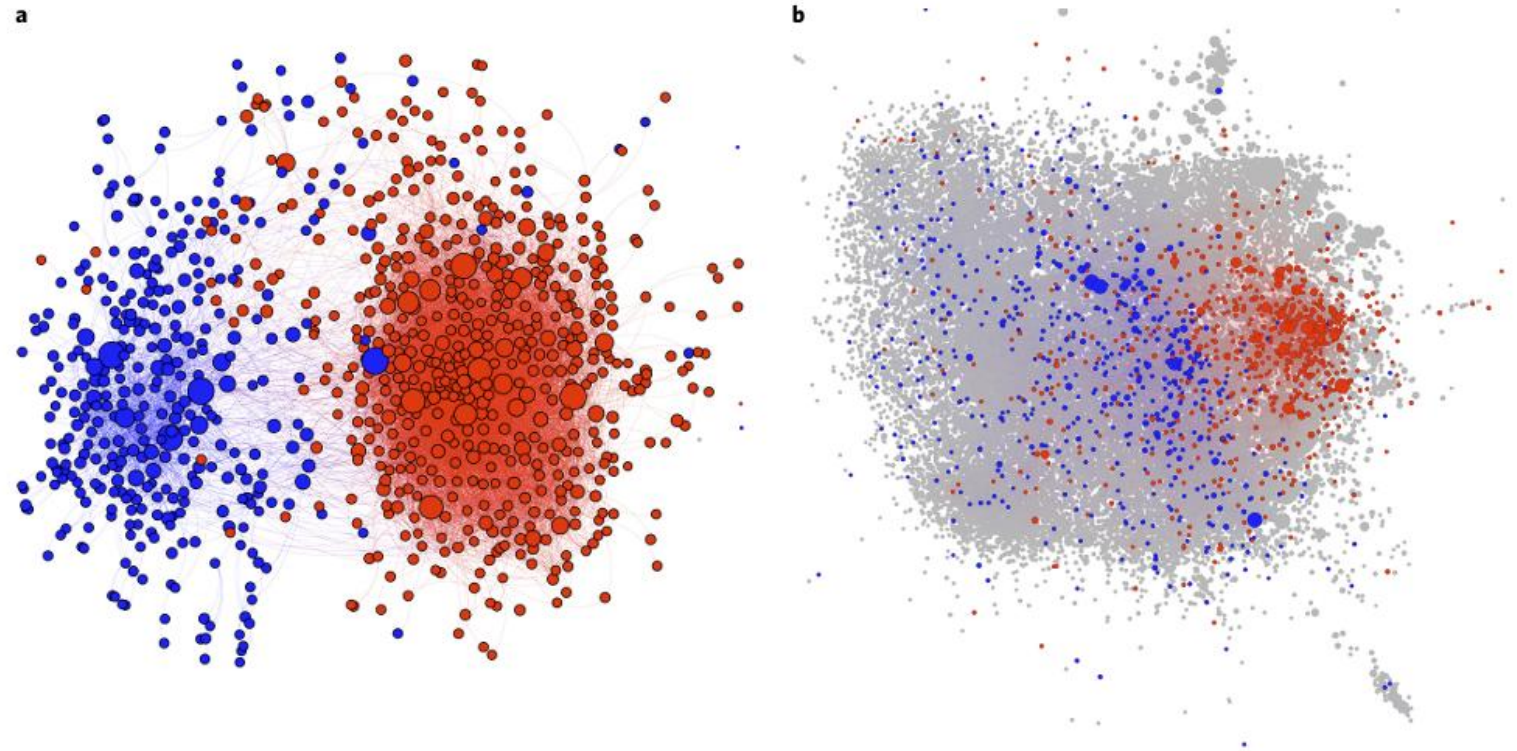


Figure 1 | Visualization of the co-purchase network among liberal, conservative and scientific books. **a**, Links between 583 liberal (blue) and 673 conservative (red) books. **b**, Links between these books and science (grey) books. As shown in **a**, 97.2% of red books linked to other reds and 93.7% of blue books linked to other blues. A small number of books were more likely to be co-purchased with books of a different colour. We subjected these blinded books to additional judges and found that the original codings were nearly all correct. A number of red 'orphans' were written by moderate Republicans critical of the religious right while blue 'orphans' were written by progressive community organizers such as Saul Alinsky (*Rules for Radicals*⁴⁰), later rediscovered by the Tea Party who reference their effective ethnic blue-collar organizing tactics. In **b**, the broader distribution of blue-linked science books indicates that readers of blue books have broader disciplinary interest and co-purchased books more centrally located in the network of co-purchased science books.

Results

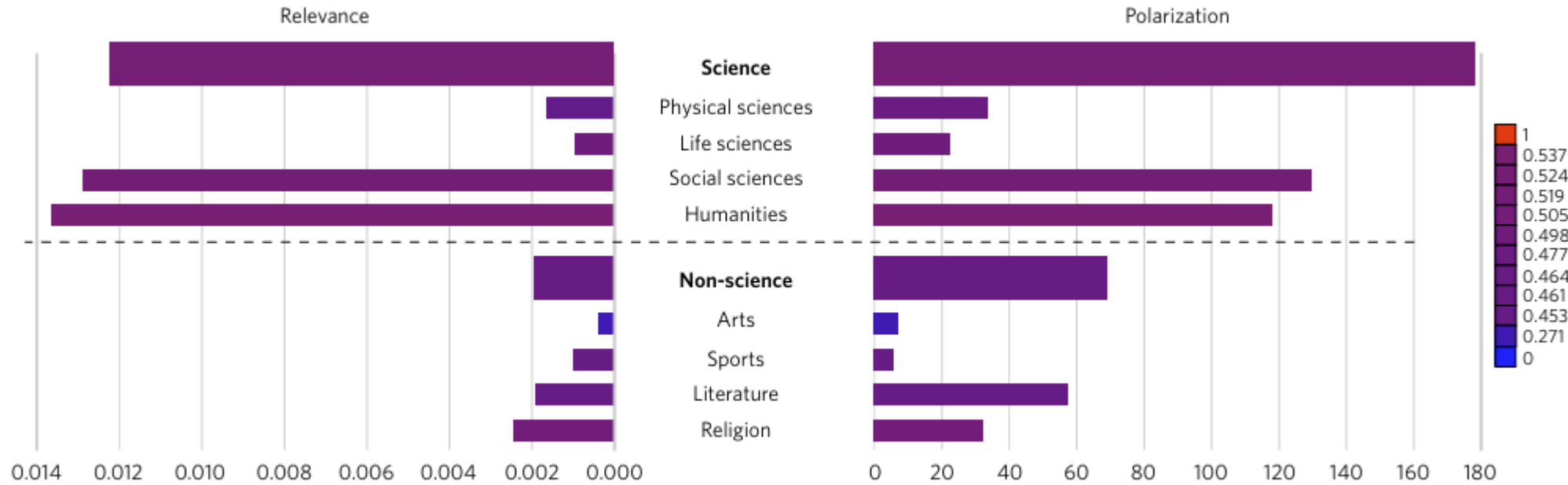


FIG 2

- Science books are more politically relevant and polarized than non-science books (humanities and social science)
- Books on the performing arts are most aligned with liberals

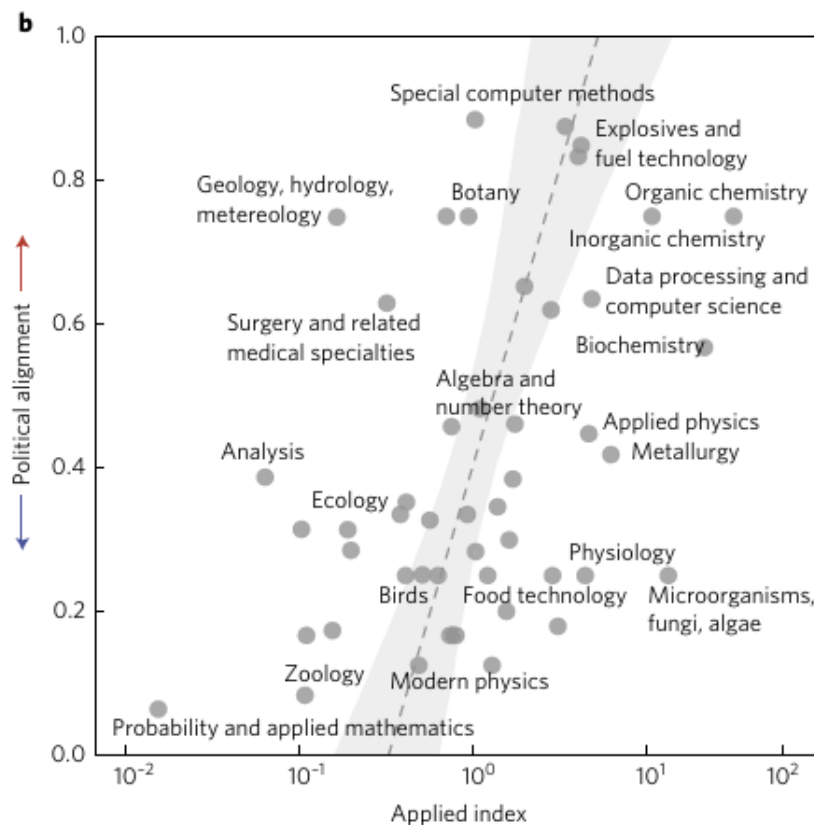
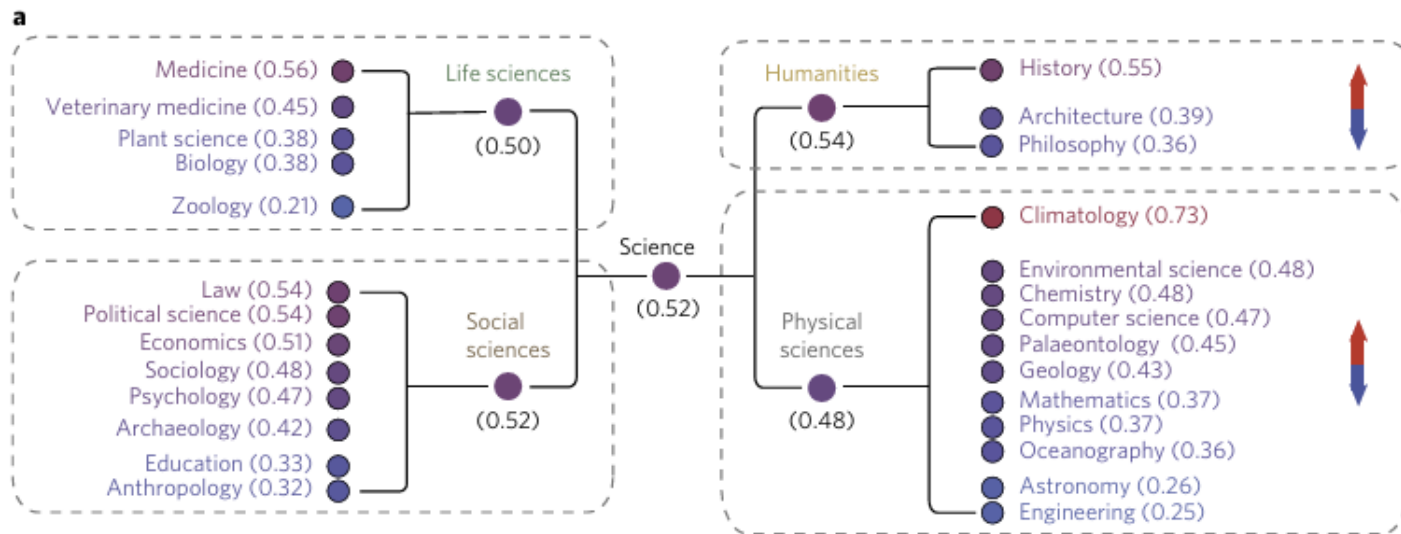
Results

FIG 3

(a) Conservatives: medicine, law, climatology

Liberals: zoology, anthropology and philosophy

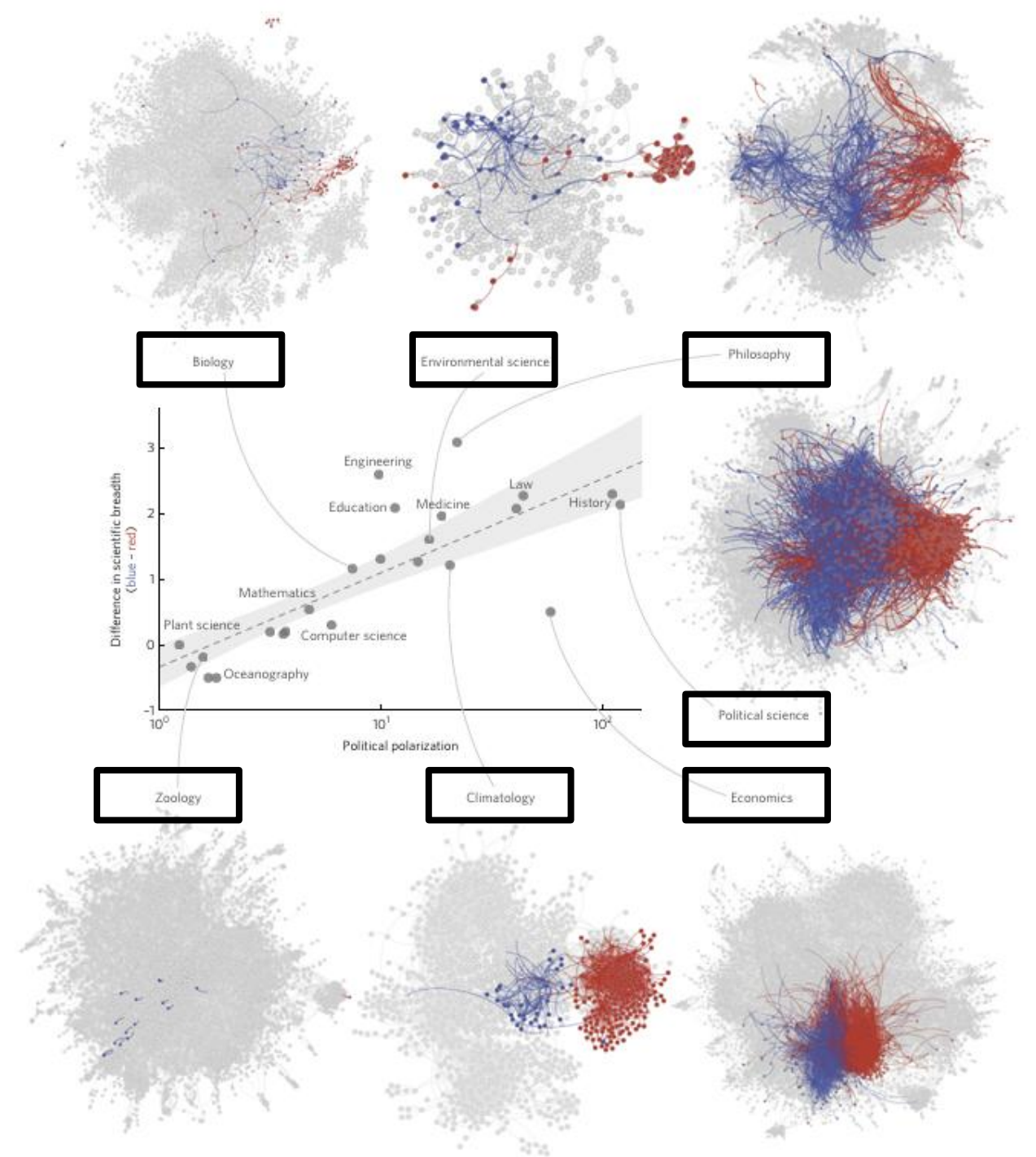
(b) Applied index predicts Conservative alignment



Results

FIG 4

- In seven disciplines: high polarization in partisan co-purchasing
- Compared with conservatives, liberals tend to purchase a more diverse set of science books (core or peripheral)
- Scatter plot: more polarized disciplines, more difference between the breadths of blue and red books



Questions

Visualization-related

- Fig. 2 was challenging to interpret, the purple shades that represent the alignment look very similar. **At what extent, researchers balance between simplicity and complexity when presenting their findings in figures?**
- Fig. 3a is creative in using gradient colors with numbers to include both accuracy and readability. **How do we determine what information to omit without unintentionally misleading the reader?**

Questions

- Figure 1-b shows that the red books appear more clustered, while the central region contains many overlaps between red and blue books. **I am curious about how this pattern should be interpreted. What interpretation might be appropriate for this zone of interaction or crossover between the two groups?**

Non visualization-related

- How other potential factors, like recommendation algorithms may work in the process of leading to co-purchases of books online? What may amplify or reduce such preference differences.